BUILD · COMPETE · GROW

HackAgon

Run your own coding contest.
We'll hold the money and keep it fair.

Advanced Java Project

Guide: Er. Ram Babu Buri

Team

Hiten Sharma (24EARCS062)
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YOUR SKILLS. OUR PROMISE.

Contests need trusted money and *fair* play

Prize money depends on trust

Student contests often run on trust: prize money sits in someone's pocket.

Honest players lose out

Cheating in online rounds goes unchecked, so honest players lose.

Luck competes with skill

Many platforms mix skill with luck draws and spin wheels.



HackAgon is skill-only. Built for students and young developers in India.

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THE PROJECT PROMISE

What the project sets out to *do*

1 **Host and join**

Let anyone host or join a coding contest.

2 **Protect the money**

Hold prize pools and entry fees in escrow; 1% fee on the pool only.

3 **Keep matches fair**

Proctoring and server-side timing keep matches fair.

4 **Judge and rank**

Sandbox judging shows a live scoreboard.

5 **Learn from mistakes**

An AI assistant helps players learn from mistakes.

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PICK YOUR FIGHT

Four ways to *play*



Tournaments

Prize pool and entry fees in escrow, fixed prize split, optional proctoring and automatic payouts.



Arena

Open rooms, mostly free.



Duel

1v1, same questions, first to pass all wins. Invite a friend or auto-match.



Learning

Every mistake saved with the fix, plus free books.

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BUILT WITH ADVANCED JAVA

Technology *stack*

Language

JAVA 17 | SERVLETS + JSP | JSTL | MAVEN WAR | NO FRAMEWORK

Frontend

HTML | CSS | VANILLA JAVASCRIPT | XML

web.xml, pom.xml, context config

Server and deploy

APACHE TOMCAT | DOCKER | RENDER

Database

TIDB CLOUD | JDBC | HIKARICP

MySQL-compatible, SSL

Services

BREVO EMAIL OTP | GROQ API AI ASSISTANT

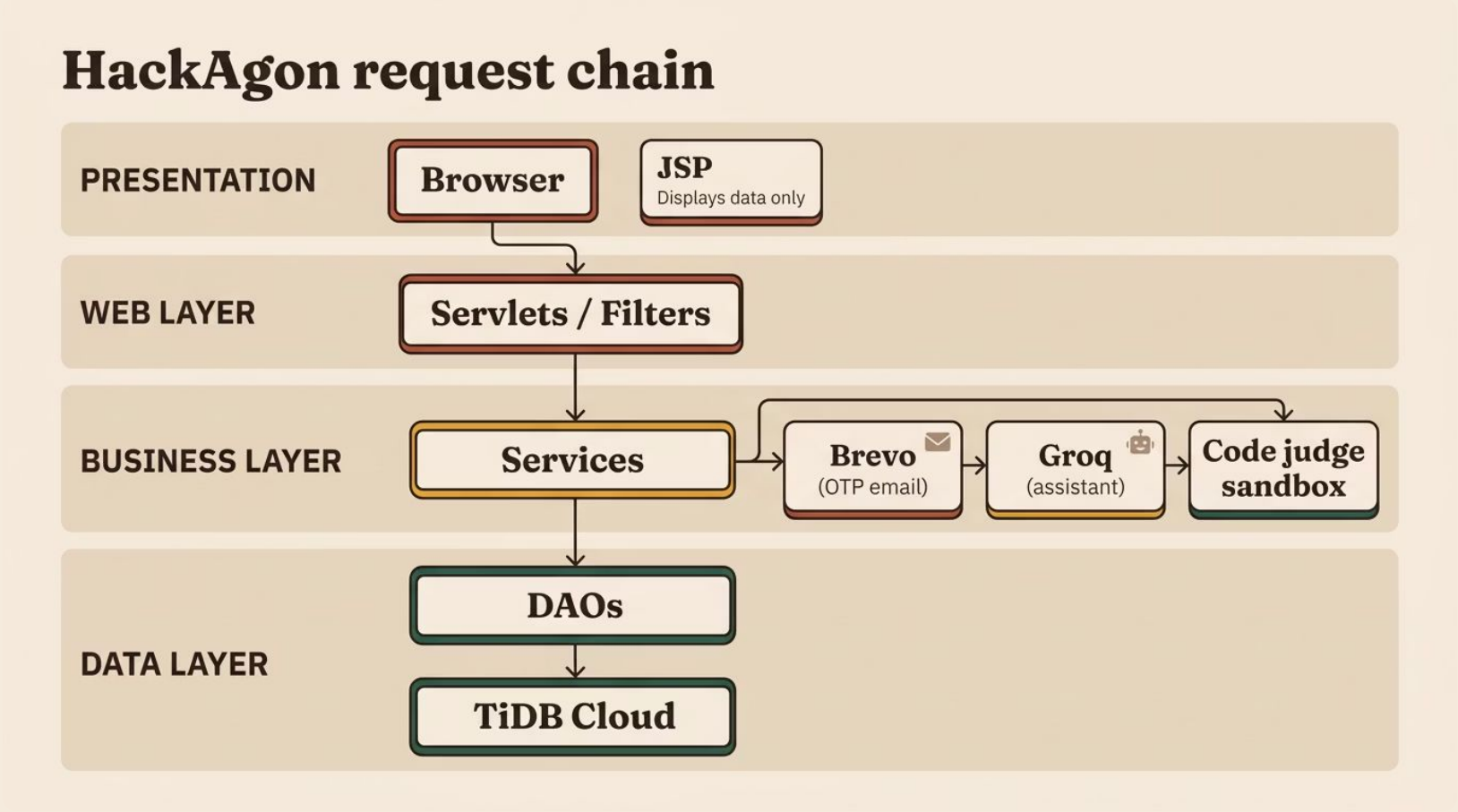
Security

BCRYPT | JWT/SESSION AUTH | CSRF FILTER | PREPAREDSTATEMENTS ONLY

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ONE RESPONSIBILITY PER LAYER

Architecture: Servlet to Service to *DAO*



Request handling, business logic and database access stay in separate layers. JSP only displays data.

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Database design: money in paise, ledger *append-only*



Money is stored as integer paise. wallet_ledger is append-only, never updated or deleted.

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GUARD EVERY ENTRY POINT

Authentication and *security*

Verify email

Email OTP via Brevo for signup and forgot password.

Protect credentials

BCrypt password hashing, JWT/session cookies.

Validate form requests

CSRF filter on every form.

Keep SQL parameterized

PreparedStatements only, so no SQL injection.

Enforce access rules

AuthFilter and RoleFilter guard every protected page.

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ONE MATCH, ONE SOURCE OF TRUTH

Match flow: from joining to final *ranks*

01	02	03	04	05	06
Join room	Solve questions	Code judged in sandbox	Store verdict and score	Live scoreboard	Final ranks

Score each solution once

Solved questions never score twice. The live scoreboard updates every 3-5 seconds.

The server owns the countdown

Changing the computer clock gains nothing.

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SIMPLE. SECURE. FAIR.

Escrow: nobody's money moves until the contest is *over*

1. **Host funds the pool**
2. **Players pay entry**

   Held in escrow with no fee on entries.
3. **Results lock**

   When the host confirms or after 24 hours with no dispute.
4. **Winners paid automatically**

   To a verified UPI ID.

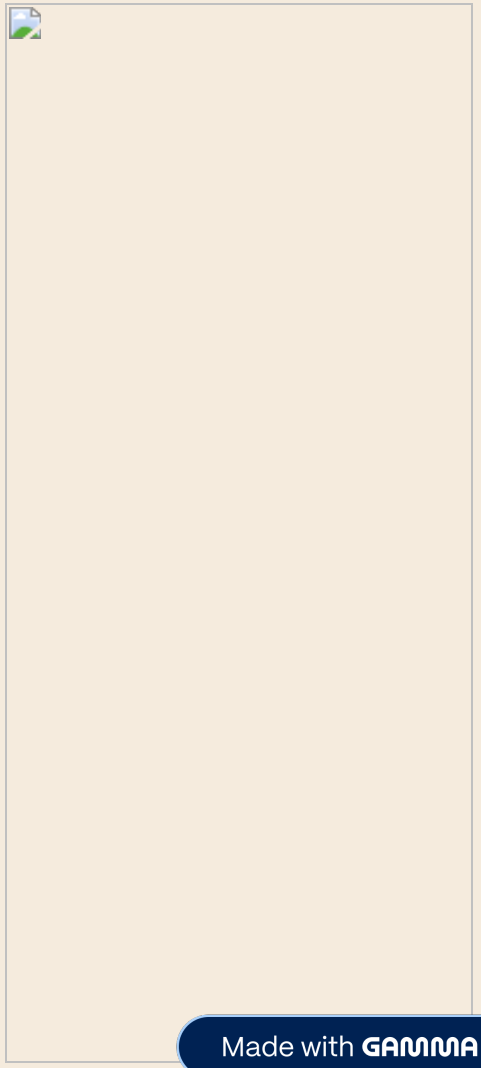
1% platform fee from the prize pool only. 100% of entry fees held in escrow.

Prize split example

Item	Share / amount
Prize pool	Rs 10,000
Platform fee (1%)	1% · Rs 100
Rank 1	50% · Rs 4,950
Rank 2	30% · Rs 2,970
Rank 3	20% · Rs 1,980

Refund rules

- Cancel before start or minimum players not met: full refund of entry fees and pool.
- A player eliminated for cheating gets no refund.



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SIMPLE. SECURE. FAIR.

Fair play, learning and what's next

Proctoring

- Camera check; webcam snapshot every 20-30 seconds, visible only to host/admin.
- Tab switch or fullscreen exit locks the match on the server, with a 30-second grace.
- Similarity check and violation log.

We stop casual cheating. We can't stop all of it.

AI assistant and Learning

- AI assistant (Groq) gives hints, explanations and help with doubts.
- Learning saves every mistake with the fix.
- Free books and achievements like Room Master support continued learning.

Deployment

Dockerized Tomcat on Render with TiDB Cloud.

Future scope

Live payments, organizations, SSE instead of polling, and load testing.

Thank you / Q&A

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